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Optional Introduction to Jupyter and R

Introduction to Generalized Linear Models

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Binomial Regression

Binomial Regression and Goodness of Fit

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Assignments: GLMs for Binomial Data

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Introduction to Generalized Linear Models

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1. During the Severe Acute Respiratory Syndrome (SARS) outbreak of 2003, some researchers believed that the treatment Ribavirin may be helpful in preventing death due to SARS. Consider a statistical model with the dosage level of Ribavirin as a continuous predictor and fatality ("Death" or "No Death") as a response. This model violates the standard linear regression assumptions. **3 / 3 points**

- (●) True
- ( ) False

Correct

2. Generalized linear models (GLMs) extend the linear regression framework to allow for non-normal responses, such as counts. **3 / 3 points**

- (●) True
- ( ) False

Correct

3. Standard linear regression is a type of generalized linear model (GLM). **3 / 3 points**

- (●) True
- ( ) False

Correct

4. The predictors of a generalized linear model (GLM) are the random component. **3 / 3 points**

- ( ) True
- (●) False

Correct

5. The link function in a generalized linear model (GLM) connects the random response to a linear combination of predictor variables. **3 / 3 points**

- (●) True
- ( ) False

Correct

6. A generalized linear model (GLM) includes: **4 / 4 points**

- [x] A link function.

Correct

- [ ] A systematic component consisting of a linear combination of continuous predictors and fixed parameters.
- [x] A systematic component consisting of a linear combination of discrete or continuous predictors, and fixed parameters.

Correct

- [ ] A systematic component consisting of a linear combination of discrete predictors and fixed parameters.
- [x] A random response component.

Correct

7. Generalized linear models require that the response comes from the exponential family of distributions. **3 / 3 points**

- (●) True
- ( ) False

Correct